TECHNIJIAN

technology as a solution

AI CONSULTING FOR PRECISION MANUFACTURING

My AI

AI consulting built for the shop floor — intelligent estimating, smart fabrication, quality intelligence, and audit-ready ITAR/AS9100 documentation. Purpose-built for precision manufacturers.

25+	ITAR	~70%	100%
YEARS EXPERIENCE	& AS9100 AWARE	FASTER QUOTES	HANDS-ON

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The Manufacturing AI Challenge

Precision manufacturers know AI could accelerate quoting, improve quality, and reduce documentation overhead — but generic AI tools don't understand engineering drawings, BOMs, or the compliance reality of ITAR and AS9100.

Quoting Bottlenecks

Estimators spend hours reading drawings, calculating material, and building BOMs. Quotes lag. Margins suffer when rushed. Reviewers miss cost drivers.

→ AI estimating intelligence & spec analysis

Production Chaos

Scheduling is manual. Machine utilization is invisible. Quality escapes happen because no one caught the pattern in time. Tribal knowledge walks out the door.

→ Smart fabrication & quality intelligence

Documentation Overhead

AS9100, ISO 9001, ITAR, and customer PPAP packages demand massive documentation. Manual paperwork eats engineering time. Audits become fire drills.

→ Automated compliance documentation

ITAR & CUI Data Risk

Cloud AI tools that send data offshore are a no-go for ITAR-controlled or CUI data. But on-prem AI has to be done right — or it creates more risk.

→ Air-gapped / US-only AI architectures

Why AI for Manufacturing Now?

- Modern AI can read engineering drawings, extract features, and generate instant BOMs — shrinking quote turnaround from days to hours
- Precision shops using AI for production scheduling and quality prediction are seeing 30-50% reductions in quality escapes and rework
- ITAR & AS9100 compliance doesn't preclude AI — it demands the right architecture: air-gapped, US-only, audit-ready, access-controlled
- Tribal knowledge loss is accelerating as senior estimators & engineers retire; AI knowledge systems preserve and scale that expertise

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Manufacturing AI Services

AI solutions purpose-built for precision manufacturing — estimating, fabrication, quality, and compliance documentation.

Estimating & Quoting Intelligence

- ✓ Drawing intake & GD&T feature extraction
- ✓ Automated BOM generation & material calcs
- ✓ Historical quote benchmarking & win-rate analysis
- ✓ Complexity scoring & margin modeling
- ✓ Customer-specific spec & revision handling
- ✓ Quote turnaround: days → hours

Smart Fabrication & Production

- ✓ Production scheduling optimization
- ✓ Machine utilization & OEE intelligence
- ✓ Work-in-progress tracking & delay prediction
- ✓ Setup time reduction via AI sequencing
- ✓ Capacity planning across multi-process shops
- ✓ Shop-floor data harmonization

Engineering & Quality AI

- ✓ Drawing/spec review with AI-assisted markup
- ✓ Historical quality escape pattern detection
- ✓ Root cause analysis with knowledge retrieval
- ✓ NCR (Non-Conformance Report) summarization
- ✓ Supplier quality scoring & risk ranking
- ✓ Engineering knowledge preservation (retirees)

Compliance & Documentation

- ✓ AS9100, ISO 9001, ITAR-compliant architectures
- ✓ PPAP, FAI, and CoC automated packaging
- ✓ Audit trail generation for every AI decision
- ✓ Air-gapped & US-only deployment options
- ✓ Controlled document retrieval with RBAC
- ✓ CUI data handling & NIST SP 800-171 alignment

How We Engage

1 Discover — Map workflows, pain points & AI opportunities

→

2 Design — Architect solutions for goals & compliance

→

3 Implement — Build, test & deploy with hands-on engineering

→

4 Operate — Operationalize with training & governance

Manufacturing Case Studies

AI engagements we've advised for precision manufacturers — from engineering & fabrication firms to ITAR/AS9100-certified shops.

ENGINEERING & MANUFACTURING

Mason West — Estimating & Engineering Knowledge AI

AI roadmap for a 50+ year ITAR/AS9100 certified engineering & manufacturing firm with 4 California offices: intelligent estimating with drawing analysis, smart fabrication scheduling, engineering knowledge preservation, spec review AI, and document intelligence.

50+ Years in biz

ITAR Certified

PRECISION SHEET METAL

Talsco (JVR Sheetmetal) — Intelligent Quoting Platform

AI advisory for a 55+ year ITAR/AS9100 precision sheet metal fabrication shop with multi-process capability: intelligent quoting with BOM automation, production scheduling AI, quality intelligence, and automated compliance documentation for PPAP/FAI packages.

55+ Years in biz

AS9100 Aerospace

CROSS-SHOP PATTERN

Drawing & Spec Intelligence at Scale

AI reads engineering drawings, extracts GD&T features, calculates material, and generates BOMs automatically — replacing manual quote buildup that consumed estimator hours per job.

Days→Hrs Per quote

GD&T Auto-extract

COMPLIANCE USE CASE

AS9100 & ITAR Documentation Automation

AI-generated PPAP, FAI, CoC and audit documentation packaged from shop-floor data — reducing documentation overhead 50%+ while improving audit readiness. All deployed air-gapped or US-only to preserve ITAR compliance.

50%+ Less doc overhead

Air-gap ITAR ready

KNOWLEDGE PRESERVATION

Engineering Knowledge AI for Retiring Senior Staff

AI knowledge system captures senior estimator and engineer expertise — capturing decades of tribal knowledge as queryable, retrievable institutional memory before retirement departures.

Persistent Knowledge base

RAG Searchable

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Engagement Models

Flexible arrangements designed to meet you where you are in your AI journey.

Fixed-Scope Projects

Defined scope, deliverables, timeline, and outcomes. Pay for results, not hours.

- ✓ AI Readiness Assessment (shop-floor + front office)
- ✓ Intelligent Quoting Platform
- ✓ Quality Intelligence Deployment
- ✓ Compliance Documentation Automation

Best for: Discrete initiatives with clear objectives

Flexible Consulting

On-demand strategic and implementation support for your manufacturing ops.

- ✓ ERP/MES integration strategy (Epicor, Global Shop, etc.)
- ✓ Air-gapped / US-only AI architecture reviews
- ✓ ITAR/CUI governance & access controls
- ✓ Shop-floor data harmonization planning

Best for: Ongoing guidance alongside your engineering team

Fractional AI Advisor

Retained monthly engagement providing AI leadership for your manufacturing operations.

- ✓ Strategic AI roadmap across shop floor & office
- ✓ Engineering / estimating team enablement
- ✓ AS9100, ITAR compliance QA for AI deployments
- ✓ Vendor oversight & technology rationalization

Best for: Building durable AI capability without hiring

Executive AI Workshops

Half-day or full-day briefings tailored to your processes and customer base.

- ✓ Manufacturing AI landscape & competitive positioning
- ✓ Live demos using your actual drawings & specs
- ✓ Opportunity assessment by process & product line
- ✓ 90-day pilot plan with measurable outcomes

Manufacturing Segments We Serve

- Aerospace & Defense
- Precision Machining
- Sheet Metal Fabrication
- ITAR / CUI Contractors
- Industrial Equipment
- Electronics & PCB Assembly
- Automotive Tier 1/2
- Medical Device Manufacturing

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Manufacturing AI Solution Map

How AI transforms precision manufacturing — quote to fabrication to compliance

Technijian My AI: Precision Manufacturing Edition

THE 4 CORE SERVICE PILLARS

Intelligent Engineering & Quoting

AI-driven analysis of technical specifications to provide instant, accurate cost estimates and automated BOM generation.

Smart Fabrication & Quality

Dynamic production scheduling and quality intelligence systems designed to minimize downtime and maximize yield.

Engineering Knowledge AI

Advanced drawing analysis and specification review tools that extract critical data from complex engineering documents.

Supply Chain Intelligence

Proactive monitoring and optimization of the manufacturing supply chain to mitigate risks and streamline procurement.

MANUFACTURING AI USE CASES

Instant Spec Analysis & BOM

The AI scans technical requirements to generate comprohensive B'R of Materials instantly, reducing the quoting cycle.

Drawing & Spec Review

Automated engineering reviews identify potential manufacturing issues early by analyzing CAD data and technical drawings.

Real-Time Production Scheduling

Smart fabrication logic continuously optimizes the floor schedule based on machine availability and quality metrics.

COMPLIANCE & REGULATORY RIGOR

ITAR & AS9100 Framework

Built-in protocols ensure all data hancling meets International Traffic in Arina Regulations and Aerospace Quality Management standards.

ISO 9001 Audit Trails

Automated documentation of every AI-assisted decision creates a contbnious, audit ready trail for quality management systems.

KEY PERFORMANCE RESULTS

Accelerated Quote Turnaround

Significant reduction in time-to-quote through automated technical analysis and historical data cross-referencing.

Minimized Quality Escapes

Enhanced quality intelligence coiches anomalies before they leave the production floor, reducing waste and rework.

Audit-Ready Documentation

100% automated record-keeping ensures the facility is always prepared for external compliance inspections.

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Ready to Put AI to Work on the Shop Floor?

Schedule a consultation to explore how AI can accelerate quoting, improve quality, and automate compliance documentation — with architectures designed for ITAR, AS9100, and ISO 9001 environments.

Schedule Consultation | **AI Briefing**

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About Technijian — Manufacturing AI Consulting

Founded in 2000 by Ravi Jain, Technijian is an AI strategy and implementation consulting firm based in Irvine, CA. We specialize in AI solutions for precision manufacturers — aerospace, defense, precision machining, sheet metal fabrication, and ITAR/CUI-controlled production. Our work combines 25+ years of technology operating experience with deep manufacturing domain expertise: engineering drawings, GD&T, BOMs, AS9100, ITAR, ERP/MES integration. We deliver the full lifecycle — from strategy and architecture through air-gapped or US-only implementation, governance, and operational rollout.

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